

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION

CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27

SET B — ADVANCED HOTS & ANALYTICAL THINKING**CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-B-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET B - Advanced HOTS & Analytical Thinking):

1. The question paper comprises **25 Higher Order Thinking Skills (HOTS) Questions** of **1 mark** each.
2. Designed to test **deep causal reasoning, critical source evaluation, constitutional mechanics, and economic analysis**.
3. All questions are compulsory. There is no negative marking for incorrect answers.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **ADVANCED HOTS INSTRUCTIONS:** Carefully analyze the socio-historical contexts, geomorphic cause-and-effect relationships, and constitutional principles before selecting your answer.

Q.1 Why do the shallow, intermediate, and deep-focus earthquakes align specifically along the Pacific 'Ring of Fire'?

- (A) The region is dominated by convergent subduction zones where dense oceanic lithosphere is thrust beneath continental plates
- (B) The Pacific Ocean contains more water weight than all other oceans combined
- (C) The earth's rotation stops completely beneath the Pacific Ocean
- (D) Undersea coral reefs generate volcanic magma explosions

Q.2 How does the travel velocity of primary seismic waves (P-waves) differ fundamentally from secondary seismic waves (S-waves) through the Earth's interior?

- (A) P-waves are longitudinal compressional waves that travel fastest and penetrate both solids and liquids; S-waves are transverse shear waves that travel only through solids
- (B) S-waves travel at the speed of light while P-waves travel at the speed of sound
- (C) P-waves cannot pass through solid rocks while S-waves pass through everything
- (D) Both wave types travel at identical speeds and have identical particle motion

Q.3 The discovery of identical fossil remains of the ancient freshwater reptile *Mesosaurus* in both eastern South America and southwestern Africa provided decisive evidence for:

- (A) Continental Drift and the historical existence of the supercontinent Pangaea/Gondwanaland
- (B) The ability of freshwater reptiles to swim across 6,000 km of open saltwater oceans
- (C) Independent parallel evolutionary creation on two isolated continents
- (D) Sudden shifting of the earth's orbit around the sun

Q.4 Why do rivers flowing through high mountainous terrain carve steep V-shaped valleys, whereas rivers traversing low alluvial plains meander in broad floodplains?

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| <p>(A) Mountain rivers possess high potential energy and steep gradients causing vigorous vertical downcutting; lowland rivers have low gradients where lateral bank erosion and sediment deposition dominate</p> | <p>(B) Mountain rivers are frozen solid while lowland rivers are made of chemical acid</p> |
| <p>(C) Lowland rivers flow uphill against gravitational forces</p> | <p>(D) Mountain rocks are softer than alluvial river clay</p> |

Q.5 During the formation of an oxbow lake, what hydraulic process causes the river to cut through the narrow neck of a meander loop?

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| <p>(A) High flood discharge increases river velocity, causing the river to take the straightest path of maximum hydraulic gradient and abandon the loop</p> | <p>(B) A sudden tectonic earthquake lifts the river channel into the air</p> |
| <p>(C) Underground magma melts the outer meander embankment</p> | <p>(D) The river permanently ceases to carry silt and water</p> |

Q.6 Why does a continental collision boundary, such as the collision between the Indian Plate and the Eurasian Plate, form towering fold mountains (Himalayas) but virtually NO active explosive volcanism?

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| <p>(A) Both colliding slabs consist of buoyant continental crust, preventing deep subduction of hydrated oceanic crust into the mantle necessary to generate volcanic magma</p> | <p>(B) The Himalayan rocks are too cold to allow volcanic eruptions</p> |
| <p>(C) Continental plates dissolve completely upon collision</p> | <p>(D) The Indian plate is moving southwards away from Asia</p> |

Q.7 What is the primary geomorphic distinction between a delta formed by a river and an estuary?

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| <p>(A) A delta is a depositional alluvial landform where low wave energy allows silt to build out seaward; an estuary is a drowned river mouth where strong marine tides scour sediments away</p> | <p>(B) A delta only forms in arctic frozen seas, while an estuary forms in tropical lakes</p> |
| <p>(C) Deltas are carved by wind erosion, while estuaries are formed by desert glaciers</p> | <p>(D) An estuary has zero interaction with ocean waters</p> |

Q.8 How does frost wedging (freeze-thaw weathering) mechanically fracture massive granite boulders in alpine mountain climates?

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| <p>(A) Water seeps into rock fissures, expands by approximately 9% in volume upon freezing into ice, and exerts immense tensile stress against joint walls</p> | <p>(B) Freezing temperatures cause granite to boil and evaporate</p> |
| <p>(C) Ice acts as a chemical solvent that dissolves silicon quartz crystals</p> | <p>(D) Snowflakes exert downward magnetic gravitational pull</p> |

Q.9 Why are crescent-shaped barchan sand dunes asymmetrical, having a gentle slope on one side and a steep slip-face on the other?

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| <p>(A) Sand grains creep gently up the windward slope, then cascade down the sheltered leeward crest at the natural angle of repose (~34 degrees)</p> | <p>(B) Underground volcanic heat repels sand grains on the leeward side</p> |
| <p>(C) Water rivers flow underneath the barchan to wash away the slip-face</p> | <p>(D) Desert animals deliberately shape the sand dune crests</p> |

Q.10 Which distinctive erosional feature distinguishes glaciated mountain valleys from non-glaciated river valleys?

- (A) Glacial valleys have flat valley floors and vertical rocky side walls (U-shaped profile with hanging valleys), whereas river valleys have interlocking spurs with V-shaped profiles
- (B) Glacial valleys are filled with molten basalt lava
- (C) River valleys never contain flowing water
- (D) Glacial valleys have circular sand dunes along their peaks

Q.11 Consider a coastal headland exposed to relentless oceanic wave attack. Arrange the sequential evolution of marine erosional landforms in correct chronological order:

- (A) Sea Cave $\rightarrow$ Sea Arch $\rightarrow$ Sea Stack $\rightarrow$ Sea Stump
- (B) Sea Stack $\rightarrow$ Sea Cave $\rightarrow$ Sea Stump $\rightarrow$ Sea Arch
- (C) Sea Arch $\rightarrow$ Sea Stump $\rightarrow$ Sea Cave $\rightarrow$ Sea Stack
- (D) Sea Stump $\rightarrow$ Sea Stack $\rightarrow$ Sea Arch $\rightarrow$ Sea Cave

Q.12 Why does chemical weathering occur at vastly higher rates in equatorial rainforests like the Amazon and Congo basins compared to polar tundras?

- (A) Chemical reactions (hydrolysis, oxidation, carbonation) require warm ambient temperatures and abundant liquid water to accelerate decomposition
- (B) Rainforests have no rocks exposed on the earth's surface
- (C) Polar tundras receive more solar insolation than equatorial forests
- (D) Rainforest trees secrete industrial acid into mountain bedrock

Q.13 What mechanism produces the distinct stratification observed in moraines versus alluvial fans?

- (A) Glacial moraines are unsorted till dumped directly by melting ice, whereas alluvial fans are sorted and layered by flowing water based on particle size and velocity
- (B) Moraines consist purely of organic plant leaves, while alluvial fans consist of iron metal
- (C) Alluvial fans are formed by desert wind gusts, while moraines form on ocean beaches
- (D) Both deposits show identical uniform sorting and fine layering

Q.14 What is the geological cause of a tsunami generated along an ocean floor subduction zone?

- (A) Sudden vertical displacement of the seafloor during an undersea megathrust earthquake, displacing thousands of cubic kilometers of the overlying water column
- (B) High summer atmospheric surface air temperatures boiling sea water
- (C) Strong seasonal trade winds blowing across the surface of the ocean
- (D) Gravitational pull of Jupiter during planetary alignment

Q.15 Why does a seismograph located in the 'seismic shadow zone' between 105 degrees and 145 degrees from an earthquake epicentre fail to detect direct P-waves?

- (A) P-waves are sharply refracted downwards as they enter the dense liquid outer core, bending away and leaving an unreached shadow ring
- (B) P-waves are completely absorbed by the earth's atmosphere
- (C) Seismic instruments within this zone are legally powered off
- (D) The earth's crust within this zone is completely elastic

Q.16 How do biological organisms contribute actively to physical weathering?

- (A) Tree roots penetrate joints and fractures in rocks, expanding as they grow and exerting mechanical wedging forces that pry rocks apart
- (B) Birds flying overhead alter atmospheric air pressure
- (C) Earthworms generate volcanic magma beneath city streets
- (D) Fish swimming in streams soften granite cliffs through breathing

Q.17 The Loess Plateau of northern China features fine yellow silt deposits exceeding 100 meters in thickness. What geomorphic process formed this massive landscape?

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| (A) Suspension transport of fine glacial and desert dust particles by strong winds from the Gobi and Taklamakan deserts over millions of years | (B) Underwater sedimentation beneath an ancient deep saltwater ocean |
| (C) Volcanic lava flows from deep undersea fissures | (D) Repeated impact of comets carrying space dust |

Q.18 What is the critical distinction between magma and lava in igneous geology?

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| (A) Magma refers to molten rock beneath the Earth's surface; lava refers to molten rock that has erupted onto the Earth's surface | (B) Magma is solid ice, while lava is liquid water |
| (C) Lava exists only in the earth's core, while magma is found on mountains | (D) There is no difference; magma and lava are different names for metamorphic rock |

Q.19 A geologist examines an ancient cliff face and finds horizontal limestone beds displaced by a vertical fracture where one block has slipped downward relative to the other. This structure is a:

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| (A) Fault line (Normal fault) | (B) Fold anticline |
| (C) Sinkhole | (D) Mushroom pedestal |

Q.20 Why do rivers flowing into calm seas (like the Bay of Bengal) develop vast bird-foot or lobate deltas, while rivers emptying into the turbulent Arabian Sea (like Narmada and Tapi) form estuaries?

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| (A) Narmada and Tapi flow through narrow rift valleys with steep gradients and swift currents, meeting high tidal ranges in the Arabian Sea that flush out sediments | (B) The Arabian Sea has no water into which rivers can empty |
| (C) Narmada and Tapi carry zero silt or rock particles throughout their course | (D) The Bay of Bengal is frozen solid throughout the year |

Q.21 What geomorphic force is responsible for shaping 'mushroom rocks' (pedestal rocks) in arid deserts?

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| (A) Wind abrasion concentrated in the lowest 1 meter of the air column, where saltating quartz grains scour softer rock bases faster than upper caps | (B) Underground gushing geysers of boiling water |
| (C) Chemical acid rain falling continuously in deserts | (D) Glacial ice caps grinding desert surfaces |

Q.22 How does chemical weathering via carbonation contribute to the development of Karst topography in limestone regions?

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| (A) Rainwater absorbs atmospheric carbon dioxide to form weak carbonic acid (H_2CO_3), which dissolves calcium carbonate (CaCO_3) along limestone joints | (B) Limestone burns into ash when exposed to sunlight |
| (C) Carbon dioxide freezes limestone into solid ice cubes | (D) Limestone turns into volcanic basalt when rain touches it |

Q.23 What is an 'anticline' versus a 'syncline' in folded geological strata?

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| (A) An anticline is an upfolded crest arch of rock strata; a syncline is a downfolded trough or basin of rock strata | (B) An anticline is an ocean trench; a syncline is a volcanic crater |
| (C) An anticline is made of salt; a syncline is made of pure coal | (D) Both terms refer to horizontal, undeformed plains |

Q.24 Why are alluvial plains formed by major river systems (like the Indo-Gangetic plain) historically the most densely populated regions on Earth?

- (A) Perennial freshwater availability, deep fertile regularly replenished soils, level terrain for transport, and ideal conditions for intensive agriculture
- (B) Total absence of rivers and complete lack of water
- (C) Permanent ice cover that preserves food without refrigeration
- (D) Rugged steep slopes that prevent humans from building houses

Q.25 Which plate tectonic boundary is exemplified by the San Andreas Fault in California?

- (A) Transform fault boundary (plates sliding horizontally past one another)
- (B) Divergent sea-floor spreading boundary
- (C) Convergent subduction mountain-building boundary
- (D) Stationary intra-plate mantle hotspot

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET B)

Q. No.	Correct	Curricular Reference & Analytical Rationale
Q.1	(A)	The Ring of Fire is ringed by oceanic subduction zones where descending tectonic plates produce intense frictional stick-slip earthquakes and explosive volcanism.
Q.2	(A)	P-waves compress and dilate rock particles in their direction of travel (passing through solids, liquids, and gases); S-waves shear particles perpendicular and cannot penetrate fluids.
Q.3	(A)	Alfred Wegener highlighted paleontological continuity: small freshwater reptiles like <i>Mesosaurus</i> could not swim across saline oceans, proving the continents were once contiguous.
Q.4	(A)	High kinetic energy and steep hydraulic gradients drive vertical corrasion in youthful upper reaches, whereas reduced velocity in mature stages shifts work to lateral corrasion.
Q.5	(A)	During flood surges, increased discharge and stream power breach the narrow neck (chute cutoff), establishing a straighter path and leaving the isolated loop as an oxbow lake.
Q.6	(A)	Continent-continent collisions involve buoyant granitic crust that resists subduction into the asthenosphere, resulting in intense crustal thickening/folding rather than volcanic arc magmatism.
Q.7	(A)	Deltas emerge when fluvial deposition outpaces marine erosion; estuaries feature powerful tidal flushing that removes river silt into deep waters without forming alluvial lobes.
Q.8	(A)	The anomalous 9% volumetric expansion of water during crystallization exerts pressures exceeding 2,000 kg/cm ² , wedging joints apart over repeated diurnal cycles.
Q.9	(A)	Saltating sand particles move up the gradual windward face (10°–15°) and avalanche down the sheltered slip-face at the critical angle of repose (~32°–34°).
Q.10	(A)	Massive glacial ice erodes both valley floor and valley walls simultaneously across its entire contact width, transforming V-shaped canyons into broad, steep-walled U-shaped troughs.
Q.11	(A)	Hydraulic wave refraction notches fractures into caves; opposing caves breach to form arches; roof collapse leaves isolated stacks; further basal abrasion reduces stacks to stumps.
Q.12	(A)	Chemical kinetics dictate that reaction rates double for every 10 °C rise in temperature; high rainfall and dense biological decay produce carbonic/humic acids that accelerate weathering.
Q.13	(A)	Glaciers deposit unsorted till ranging from massive boulders to fine flour without hydraulic sorting; flowing water deposits sediments graded by decreasing stream competency.
Q.14	(A)	Megathrust fault rupture abruptly uplifts or down-drops the oceanic crust, transferring kinetic energy to the water column and propagating open-ocean waves at jet speeds (~800 km/h).
Q.15	(A)	The sharp velocity drop at the mantle-core boundary refracts P-waves inward, creating an angular shadow zone between 105° and 145° from the earthquake focus.
Q.16	(A)	Root wedging is a potent biophysical agent: growing root systems exert lateral pressures up to 15 bars, splitting bedrock along pre-existing joints.
Q.17	(A)	Loess is wind-deposited (aeolian) dust transported in atmospheric suspension from arid deserts and peri-glacial plains and trapped by semi-arid vegetation.
Q.18	(A)	Magma is subsurface molten material charged with dissolved gases; upon breaching the surface through vents or fissures, it degasses and is termed lava.
Q.19	(A)	Faulting involves fracture and shear displacement of rock strata along a plane under tensional or compressional tectonic stresses.

Q. No.	Correct	Curricular Reference & Analytical Rationale
Q.20	(A)	Narmada and Tapi flow through hard fault-trough rift valleys with low silt loads, meeting strong Arabian Sea tidal currents that scour river mouths into deep estuaries.
Q.21	(A)	Wind-driven saltation carries dense sand grains within 1 meter of the desert floor, focusing abrasive corrosion on the rock's lower stalk.
Q.22	(A)	Carbonic acid in rainwater reacts with insoluble calcite (CaCO_3) to form soluble calcium bicarbonate ($\text{Ca}(\text{HCO}_3)_2$), excavating subterranean caves and sinkholes.
Q.23	(A)	Compressional tectonic forces fold horizontal strata into convex-upward arches (anticlines) and concave-downward troughs (synclines).
Q.24	(A)	Alluvial floodplains provide nutrient-rich silt, high water tables, navigable channels, and flat topography, sustaining dense agricultural civilisations.
Q.25	(A)	The San Andreas Fault is a right-lateral transform fault where the Pacific Plate slides horizontally past the North American Plate without creating or destroying crust.